IEEE 14-Bus GFL↔GFM Switching with Project AGSI++

Full-state implementation and present evidence

4 August 2026

Abstract

The study uses the configured IEEE 14-bus case, device parameters, and event chronology; switching remains the project AGSI++. The study profile uses a six-state operational EMF6 SG and 12-state GFL/GFM switching supersets. The decision index is saturated to $[0, 1]$. The required publication horizon is 160 s; no shorter verification gate or failed trajectory is presented as that result. The exact run exits the model-validity domain at 36.040 s after the SG trip, before the later scheduled events. Its raw accepted prefix is shown only as diagnostic switching evidence. No Bayesian-optimisation controller or altered numerical result is introduced. The requested measurement-noise overlay is seeded, presentation-only, explicitly labelled, and kept separate from switching decisions and the saved raw trajectory.

1 Case, index, and switching

The 100-MVA, 60-Hz IEEE 14-bus case has one SG at bus 1 and IBRs at buses 2, 3, 6, and 8. The configured apparent-power dispatch is mapped to P/Q with a documented common angle (`PROJECT_DERIVED`); all device values are read from the source table. The project index is

$$J_V = \frac{|V - V_{ref}|}{0.10}, \quad J_f = \frac{|f - 60|}{0.50}, \quad J_R = \frac{|\mathrm{d}f/\mathrm{d}t|}{1.0}, \quad J_P = \frac{|P_{ref} - P|}{0.20},$$

$$J_{SCR} = \max(0, 3/SCR - 1), \quad J_{lock} = \frac{|v_q|}{0.10}, \quad J_{GRA} = 1 - GRA,$$

$$\mathrm{AGSI}^{++} = \min\left(1, \max\left(0, \frac{1}{7}\sum J_k\right)\right).$$

Each IBR has its own timer. It commits GFL→GFM above $\Gamma_{on} = 0.65$ for $T_{d,on} = 0.10$ s and GFM→GFL below $\Gamma_{off} = 0.35$ with $GRA = 1$ for $T_{d,off} = 1.00$ s. The action is local, not broadcast: each IBR owns its index and timer. Coordinated SG-reference handback is separate from index-driven switching. A future coordinated controller will be selected with the project advisor; no Bayesian-optimisation block is used here.

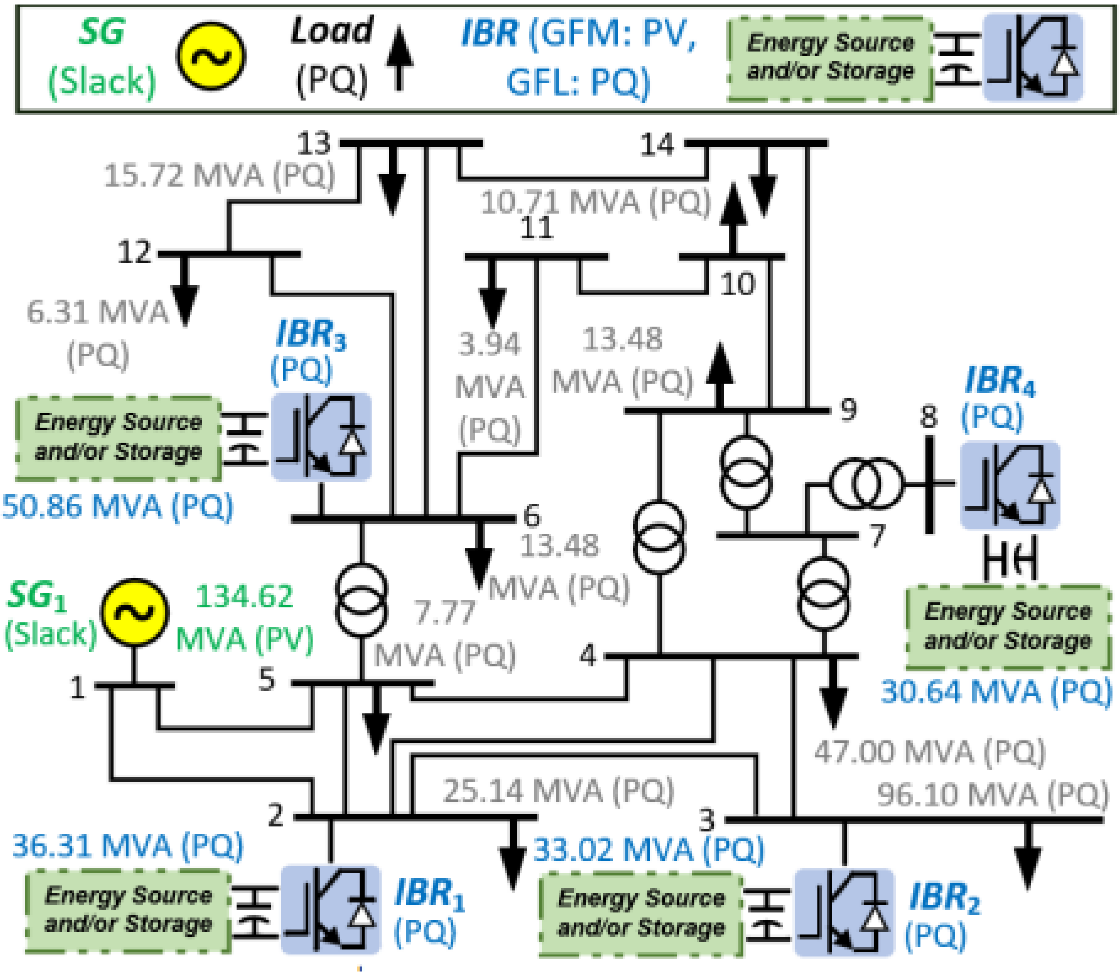

Figure 1: Network and resource locations used by the simulation.

2 Dynamic model

The SG state is $x_{SG} = [\delta, \Delta\omega, E'_q, E'_d, E''_q, E''_d]^T$ with the operational Kundur/EMF6 stator and flux equations. T_m and E_{fd} are equilibrium controls held constant; governor, AVR, synchronizer, and breaker transients are not claimed. The implemented core is

$$\dot{\delta} = \omega_0 \Delta\omega, \quad 2H\dot{\Delta\omega} = T_m - T_e - D\Delta\omega,$$

$$T'_{d0}\dot{E}'_q = E_{fd} + c_d E''_q - d_d E'_q, \quad T'_{q0}\dot{E}'_d = c_q E''_d - d_q E'_d,$$

$$T''_{d0}\dot{E}''_q = E'_q - E''_q - (X'_d - X''_d)I_d, \quad T''_{q0}\dot{E}''_d = E'_d - E''_d + (X'_q - X''_q)I_q.$$

The GFL superset contains filter current, DC-link, PLL, P/Q PI, current PI, and command-delay states. GFM contains filter current, DC-link, virtual angle/speed/EMF, voltage PI, current PI, and delay states. Source values include $L_f = 0.15$, $R_f = 0.015$, $C_{dc} = 0.10$, $I_{max} = 1.2$, $T_d = 0.02$, $k_{p,Idq} = 0.30$, and $k_{i,Idq} = 4.00$. The paper does not publish the energy-source I_{dc} law, so the implementation closes that port with ideal instantaneous power balance; this preserves V_{dc} but cannot establish long-duration energy headroom. With $V_{dq} = Ve^{-j\theta} = v_d + jv_q$, the common filter/DC plant is

$$\frac{L_f}{\omega_b}\dot{i}_d = v_{cd,del} - v_d - R_f i_d + \omega L_f i_q, \quad \frac{L_f}{\omega_b}\dot{i}_q = v_{cq,del} - v_q - R_f i_q - \omega L_f i_d,$$

$$C_{dc}\dot{V}_{dc} = I_{dc} - P_{ac}/V_{dc}, \qquad T_d\dot{v}_{c,del} = v_c^* - v_{c,del}.$$

GFL uses the PLL and P/Q PI states; GFM uses $\dot{\theta} = \omega_b(\omega - 1)$, $M\dot{\omega} = P_{ref} - P - D_v(\omega - 1)$, and $\tau_E\dot{E} = k_Q(Q_{ref} - Q) - k_E(E - |V|)$. Both modes use current error $e_{dq} = i_{dq}^* - i_{dq}$, radial I_{max} limiting, and anti-windup.

3 Power-flow operating point

Table 1: Bus power-flow results (pu)

Bus	Type	$\lvert V \rvert$	θ (deg)	P_g	Q_g	P_L	Q_L
1	REF	1.06000	0.0000	1.31895	-0.26951	0.00000	0.00000
2	PQ	1.05456	-2.9058	0.31859	0.17419	0.21700	0.12700
3	PQ	1.03301	-7.6873	0.28972	0.15841	0.94200	0.19000
4	PQ	1.04895	-5.9618	0.00000	0.00000	0.47800	-0.03900
5	PQ	1.05072	-4.9102	0.00000	0.00000	0.07600	0.01600
6	PQ	1.12705	-5.9341	0.44625	0.24399	0.11200	0.07500
7	PQ	1.09514	-6.1353	0.00000	0.00000	0.00000	0.00000
8	PQ	1.11749	-3.9176	0.26884	0.14699	0.00000	0.00000
9	PQ	1.09398	-7.6393	0.00000	0.00000	0.29500	0.16600
10	PQ	1.09237	-7.5903	0.00000	0.00000	0.09000	0.05800
11	PQ	1.10580	-6.8756	0.00000	0.00000	0.03500	0.01800
12	PQ	1.11174	-6.7892	0.00000	0.00000	0.06100	0.01600
13	PQ	1.10530	-6.9206	0.00000	0.00000	0.13500	0.05800
14	PQ	1.08161	-8.2396	0.00000	0.00000	0.14900	0.05000

Table 2: From-end line flows and losses (pu)

Line	From	To	P_{from}	Q_{from}	P_{loss}	Q_{loss}
1	1	2	0.900928	-0.203057	0.014518	-0.014696
2	1	5	0.418019	-0.066451	0.008475	-0.019813
3	2	3	0.464318	-0.000621	0.009133	-0.009246
4	2	4	0.314334	-0.079982	0.005358	-0.021354
5	2	5	0.209348	-0.060569	0.002332	-0.031219
6	3	4	-0.197092	-0.022968	0.002456	-0.007604
7	4	5	-0.447907	0.102332	0.002561	0.008079
8	4	7	0.017005	-0.115901	0.000000	0.002495
9	4	9	0.062330	-0.021423	0.000000	0.002062
10	5	6	0.090091	0.002266	0.000000	0.001610
11	6	11	0.130988	0.058687	0.001540	0.003226
12	6	12	0.085939	0.026725	0.000784	0.001631
13	6	13	0.207419	0.084233	0.002610	0.005140
14	7	8	-0.268840	-0.133746	0.000000	0.013242
15	7	9	0.285846	0.015351	0.000000	0.007516
16	9	10	-0.003741	0.022196	0.000013	0.000036
17	9	14	0.056917	0.023544	0.000403	0.000857
18	10	11	-0.093754	-0.035839	0.000693	0.001622
19	12	13	0.024155	0.009094	0.000119	0.000108
20	13	14	0.093845	0.030080	0.001359	0.002767

4 Declared 160-s study and publication gate

Time (s)	Declared event	Publication status
0–20	base operation, all IBRs initially GFL	operating point verified
20	SG1 trip	reached; all four IBRs commit GFM
36.040	state-domain validity exit	diagnostic prefix ends; later events not reached
50	every configured P_j^{load}, Q_j^{load} increased by 20%	not reached
85–85.15	temporary three-phase bus-9 fault	not reached
110	line 6–13 trip	not reached
145	SG1 and line 6–13 reclose; all loads restored to base	not reached
160	required simulation endpoint	not reached; recovery unproven

Existing tests separately cover no-event, light/severe faults, load steps, SG trip, and SG reclose. A 4-s compressed reclose case remains an internal software gate only and is deliberately omitted. The figures below are the accepted prefix of the exact run, not a 160-s recovery claim.

5 Switching and electrical diagnostic evidence

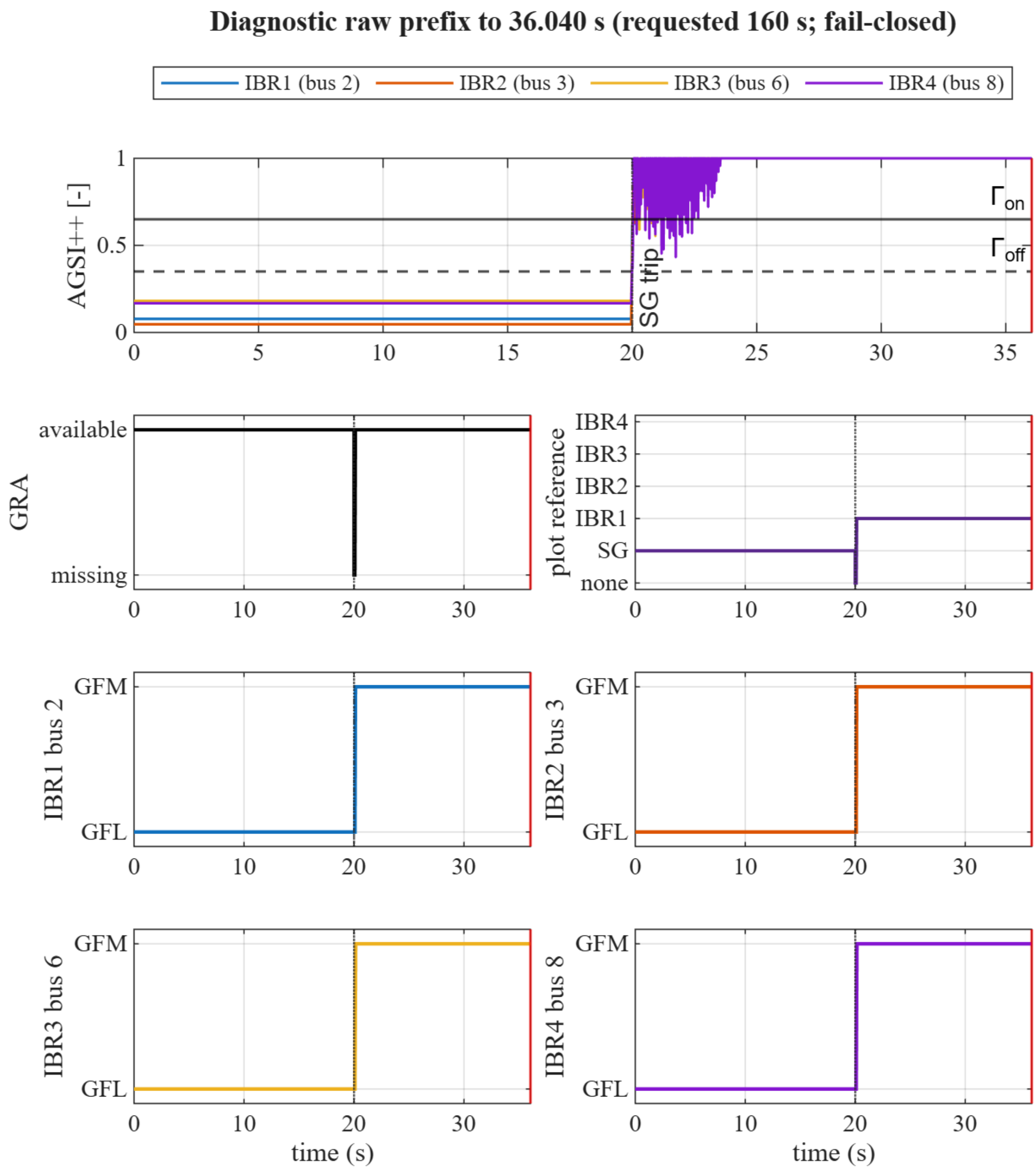

Figure 2: AGSI++, GRA, elected reference, and separate IBR1–IBR4 mode timelines. All four IBRs cross Γ_{on} after the SG trip and commit GFM locally. The run is fail-closed at 36.040 s; events at 50 s and later are therefore not claimed.

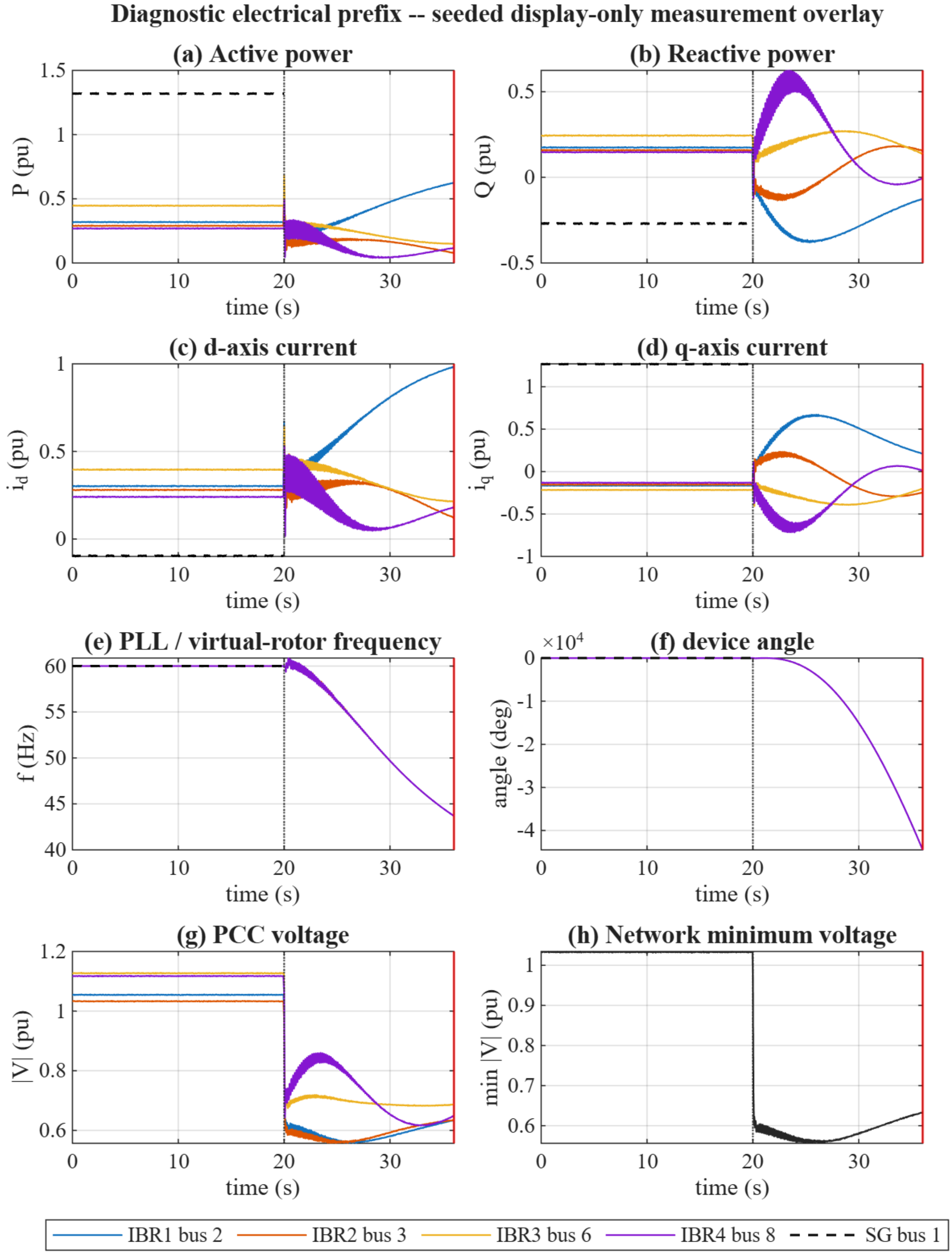

Figure 3: Diagnostic IBR and SG P, Q, i_d, i_q, f, angle, PCC voltage, and network-minimum-voltage traces. The thin seeded synthetic measurement overlay is visual only; solver states, AGSI, timers, mode decisions, and the saved raw data are unchanged. The frequency and angle drift expose the missing long-duration dispatch/energy coordination rather than recovery.

6 Reproduction

```
pf_init_paths;
```

```
addpath(fullfile('scripts','reporting'));
generate_ieee14_switch_report_figures;
```

The generator reproduces the PF tables and the exact-run diagnostic prefix. A complete 160-s recovery result remains fail-closed pending the advisor-approved coordinated-control contract. All solvers are project-owned MATLAB.